

DA-01-T3 — Behavioural Feature Specification

DiscountMate | DataBytes | Trimester 2 2026 | Feature 1: Behavioural Analytics & Persona Segmentation

Ticket ID	DA-01-T3
Priority / Level / SP	P0 / L2 / 3 SP
Planned window	W4–W5 (T2 2026)
Prepared by	Robin Le (Cross-Team Lead, DA/ML)
Reviewers	Shubhpreet Kaur (Co-Lead, Feature 1); Amoda Senarath (Co-Lead); Senushi (Feature 2 sub-lead)
Depends on	DA-01-T1 (GA4 Event Audit — Noel); DA-01-T2 (T1 Persona Review — Carlin)
Feeds into	DA-01-T4 (extraction); DA-01-T5 (cleaning); DA-01-T7 (data dictionary); DA-01-T9 (clustering); DA-01-T11 (persona interpretation)
Downstream consumers	AI/DL DL-05 (recommender); DA-02-T4 (dashboards); DA-03-T12 (persona-aware scoring, P2)
Status	Draft v1.0

1. Executive Summary

DiscountMate is a comparison platform, not a retailer. Users come to decide — not to buy from us. Behavioural segmentation should therefore describe how users make comparison decisions, not how much they spend with us.

This specification defines the behavioural feature set, data requirements, and clustering approach that will translate GA4 event data into behaviour-derived user segments — replacing the T1 rule-based persona framework as this trimester's primary segmentation output. It applies the **Recency, Frequency, Engagement (RFE) framework**, the accepted adaptation of RFM for comparison and media platforms where value is derived through engagement rather than transaction (Smith, 2023).

T2 deliverable: 13 behavioural features across 4 dimensions, buildable from GA4 events currently firing on production. Two clustering algorithms (K-means and DBSCAN) run in parallel. Segment definitions mapped against the five T1 personas.

Beyond T2: The specification identifies which behavioural signals require frontend instrumentation, platform functionality, or catalogue data that does not yet exist. Schemas for these deferred features are documented so they can be added to future iterations without redefining the model.

2. Business Context

What DiscountMate does

DiscountMate helps Australian shoppers compare grocery prices across four retailers — Coles, Woolworths, Aldi, and IGA. A user visits DiscountMate, compares products or specials, and — when they decide — leaves DiscountMate to complete the purchase at the retailer of their choice. The platform's value is measured by whether it helps users make better comparison decisions, not by transaction volume on our own site.

Why segmentation matters this trimester

The T1 team defined five self-declared personas — Budget-Conscious Shopper, Health Enthusiast, Family Planner, Convenience Seeker, and Premium Shopper — assigned via a questionnaire and static rules. This gave the team a shared vocabulary and a starting point, but the personas describe what users say they want, not how they behave. The T2 team objective is to close that gap by deriving personas from observed GA4 behaviour, then comparing the two views.

Who consumes the output

Behaviour-derived segments feed three downstream workstreams. This spec is the contract those workstreams depend on.

Downstream consumer	Uses the segments for	Needed by
AI/DL — DL-05 Personalised Recommendation Model	Persona → product mapping for recommender interface	W9
DA-02 — Insight Surfacing (Senushi)	Persona visualisation in stakeholder Power BI dashboards	W8
DA-03 — True Value Promotion V2 (Amoda)	Optional per-persona scoring weights (P2 extension)	W9–W10

The output is not consumed by end users directly. It is an internal artefact that shapes what other DiscountMate features do.

3. Objectives and Non-Goals

In scope for this specification:

- Define the behavioural feature list — variables, sources, definitions.
- Specify preprocessing and data-quality requirements for the extraction team.
- Set the clustering approach — algorithms, distance metric, cluster-count guidance.
- Establish the mapping approach for comparing behavioural clusters against the T1 personas.

Out of scope for this specification:

- The clustering model implementation — sits with DA-01-T9 (Shubhpreet, W7–W8).
- The extraction SQL — sits with DA-01-T4 (W5-6).
- Dashboard design consuming these segments — sits with DA-02 (Senushi).
- The recommender interface consuming persona output — sits with AI/DL DL-05 (Ethan, W9).

This document is a contract, not an implementation. It tells the extraction, clustering, and downstream teams what they will receive and what they will not.

4. Approach — Why RFE for a Comparison Platform

We adapt the Recency, Frequency, Engagement (RFE) framework because DiscountMate does not sell to users — it helps them decide where to buy. Classical RFM analysis is designed for platforms that capture value through transactions, where M (monetary value) is the outcome metric. For platforms where value is captured through engagement rather than transaction, consumer analytics literature substitutes E (Engagement) for M.

Smith (2023) states this directly: for apps and comparison platforms, "patronage is essentially equivalent to usage." DiscountMate sits in this category.

Our four behavioural dimensions map to R + F + E, plus one platform-specific extension that captures how users navigate through the site.

Dimension	Framework element	Business question it answers
A. Engagement Intensity	Frequency + Engagement	How much of the user's attention are we getting?
B. Feature Affinity	Engagement (revealed preference)	What is the user actually here to do — compare, hunt specials, plan lists, browse categories?
C. Exploration Style	Engagement (behaviour shape)	Are they task-directed (know what they want) or exploratory (discovering)?
D. Recency & Tenure	Recency	Are they a returning user, or an occasional / new visitor?

Dimension B is where the DiscountMate-specific signal lives. A user who spends 60% of their pageviews on /compare is behaviourally different from a user who spends 60% on /specials, even if both have identical session counts and engagement times. The feature-affinity ratios convert URL patterns into the signal that separates comparers from deal-hunters from list-planners.

Attitude versus behaviour

The T1 personas come from self-report — users describing themselves through a questionnaire. Consumer research consistently finds gaps between what people say and what they do; Smith (2023) refers to this as the *attitude-behaviour gap*. We should expect our behaviour-derived clusters to diverge from T1 personas in places. Where they agree, the T1 framework is validated for that segment. Where they split a T1 persona into sub-groups, the behavioural data has surfaced nuance. Where they cross T1 boundaries entirely, the T1 questionnaire may be asking the wrong question. All three outcomes are informative — none is a failure of the model.

5. Behavioural Feature Specification

13 features across 4 dimensions. All are derived from GA4 events currently firing on production, using URL patterns (event_params.page_location) and event parameters (engagement_time_msec, percent_scrolled, page_referrer) rather than custom event names. This makes the feature set robust to the custom-event instrumentation gap currently affecting the production tag.

5.1 Dimension A — Engagement Intensity

Feature	Definition	Source	Type
sessions_count	Distinct sessions per user in the reporting window	session_start	Integer
avg_engagement_time_sec	Mean engagement time per session	engagement_time_msec param	Float

Feature	Definition	Source	Type
pageviews_per_session	Total pageviews divided by session count	page_view, session_start	Float
avg_percent_scrolled	Mean scroll depth per pageview	percent_scrolled param	Float 0–100

5.2 Dimension B — Feature Affinity

Ratios are used rather than raw counts so that low- and high-activity users are directly comparable.

Denominator is the user's *total pageviews* within the reporting window, excluding pageviews on non-production hostnames (see §6.2).

Feature	Definition	Type
compare_ratio	Pageviews on /compare divided by total pageviews	Float 0–1
specials_ratio	Pageviews on /specials divided by total pageviews	Float 0–1
mylists_ratio	Pageviews on /my-lists divided by total pageviews	Float 0–1
category_browse_ratio	Pageviews on /category/{id} divided by total pageviews	Float 0–1
product_detail_ratio	Pageviews on /product/{id} divided by total pageviews	Float 0–1

5.3 Dimension C — Exploration Style

Feature	Definition	Type
unique_pages_visited	Count of distinct page_location values across the user's sessions	Integer
browsing_entropy	Simpson diversity index over page_referrer chains within the DiscountMate domain — captures variety of in-app navigation paths	Float 0–1

The browsing_entropy feature is grounded in the systematic-vs-unsystematic behaviour concept in Smith (2023): users who move through a small set of pages repeatedly score low (systematic, task-directed); users who navigate widely score high (unsystematic, exploratory).

5.4 Dimension D — Recency and Tenure

Feature	Definition	Type
days_since_first_visit	Days between the reporting-window end and the user's first_visit timestamp (tenure)	Integer
days_since_last_activity	Days between the reporting-window end and the user's most recent session_start (recency)	Integer

6. Data Requirements

6.1 Inputs

- Primary source: GA4 raw event export via BigQuery — Events_YYYYMMDD and pseudonymous_users_YYYYMMDD tables from the DiscountMate-GA property (project 459801038998).
- Grain: one row per GA4 event; multiple events per session; multiple sessions per user_pseudo_id.
- Reporting window: rolling 14 days for the W6 validation run (since mid-trimester break); extended to the full T2 window for the W7–W8 final clustering run.

6.2 Data cleaning (executed by DA-01-T5)

Six mandatory filters and normalisations applied before feature aggregation. These are requirements, not suggestions — the DA-01-T5 team should treat this section as the **acceptance criteria** for their sub-task.

1. Hostname filter. Retain events where device.web_info.hostname = 'discountmate.app'. Excludes localhost (developer testing) and discountmate-prod.web.app (staging), both of which currently pollute the production event stream.

2. Query-string normalisation. Strip query parameters from page_location before URL pattern matching. Removes `?gtm\_debug=..., ?create=1`, and similar debug or state parameters that inflate distinct-URL counts.

3. Bot / bounce filter. Exclude sessions containing only a single event.

4. User inclusion threshold. Include users with *at least one* session containing at least 10 seconds of engagement time. Excludes accidental loads and one-page abandonments.

5. Missing-data handling. Impute 0 for count- and ratio-based features when the underlying event was not fired. Drop the user entirely if total pageviews = 0 (nothing to cluster on).

6. Outlier capping. Cap continuous features at the 99th percentile per feature before scaling. K-means is sensitive to extreme values; capping preserves signal without letting individual users dominate cluster centroids.

6.3 Outputs — data contract for downstream teams

The cleaned, feature-engineered dataset is the interface DA-01-T3 owes to T4 → T9 → T11, and by extension to DA-02 and AI/DL. One row per included user.

Field	Type	Notes
user_pseudo_id	String	GA4 client identifier; primary key
reporting_window_start	Date	Start of the aggregation window
reporting_window_end	Date	End of the aggregation window
13 behavioural features	Numeric	Per §5.1–§5.4
is_included_in_run	Binary	User meets §6.2 inclusion criteria

- Format: CSV acceptable during development.
- Refresh cadence: weekly during model development. Automated refresh once repointed to the DE gold layer in later trimesters.

7. Clustering Methodology (for T9)

Two algorithms, run in parallel

K-means and DBSCAN answer different questions and both are useful for T11 interpretation:

- K-means asks: if we forced users into k coherent groups, what would those groups look like? Setting k around 5 lets us compare cluster centres directly against the five T1 personas. This is the interpretability-friendly view.
- DBSCAN asks: are there natural dense groupings in the data, and which users don't fit any cluster? DBSCAN's ability to label users as noise prevents us from forcing every user into a persona box, and can reveal segments K-means would miss.

Scaling

- Z-score standardisation before K-means (K-means is scale-sensitive).
- Min-max normalisation to [0, 1] before DBSCAN (makes the ϵ neighbourhood parameter interpretable across features).
- Retain an unnormalised copy for T11 cluster interpretation.

Distance metric

Euclidean at launch for both algorithms. If Dimension B ratio features distort results (a known artefact of high-dimensional ratio data), a cosine variant will be evaluated during T9.

k selection

Anchor around $k = 5$ for direct T1 persona comparability. Use elbow method and silhouette score to select the final k. Do not constrain to 5 — a smaller or larger cluster count that better fits the data is a valid outcome.

Reproducibility

Fix random seeds. Save all hyperparameters and the cluster-assignment table alongside the model artefact. This is a delivery requirement, not a nice-to-have.

Interpretability requirement

Every cluster must be describable in plain-language dimension terms — for example, "high compare_ratio and high specials_ratio, low mylists_ratio" — for T11 to convert into a named persona. If a cluster cannot be described this way, it should not be released as a persona.

8. Interpretation and Mapping to T1 Personas (for T11)

T11 will compare behaviour-derived clusters against the T1 self-declared personas. Three outcomes are possible and all are informative:

Confirmation — cluster behaviour matches a T1 persona; the T1 framework is **validated** for that segment.

Split — a T1 persona breaks into two or more behavioural sub-groups; the behavioural data **adds nuance** to the rule-based baseline.

Divergence — clusters cross T1 persona boundaries; suggests the T1 questionnaire **may not capture** that group's actual behaviour.

The table below is the hypothesis for T11 to test, not a rule for pre-assigning users to clusters. Its purpose is to give T11 an anchor for interpretation.

T1 Persona	Expected behavioural signature
Budget-Conscious Shopper	High <i>specials_ratio</i> , high <i>compare_ratio</i>
Family Planner	High <i>mylists_ratio</i> , high <i>sessions_count</i> , returning <i>tenure</i>
Convenience Seeker	Low <i>avg_engagement_time_sec</i> , high <i>product_detail_ratio</i> (goes direct to what they want)
Premium Shopper	Low <i>specials_ratio</i> , moderate <i>product_detail_ratio</i>
Health Enthusiast	High <i>category_browse_ratio</i> (interpretable once category IDs are mapped to human-readable category names — see §9.2)

9. T2 Deliverable Scope and Future Direction

9.1 In scope for T2

Everything defined in §5 through §8 is deliverable within T2 using GA4 events currently firing on production. No frontend instrumentation changes are required to produce the T2 clustering output. The 13-feature model, the two-algorithm clustering approach, and the T1 persona mapping constitute the T2 commitment.

Data volume mitigation

Real user volume is at the lower bound of what is meaningful for K-means and DBSCAN. The team will broaden the dataset in three complementary ways during the W5–W8 collection window:

- Invite directors, mentors, and cross-team members to use DiscountMate as real shoppers from W6. Aligned with the roadmap Risk 1 mitigation.
- Coordinate teammate walk-throughs enacting the five T1 persona behaviours during W6–W7. This generates labelled behavioural traces that serve as a ground-truth reference for T11 cluster interpretation, in addition to broadening the sample.
- Continue passive data collection throughout the mid-semester break and W6–W8 to maximise the sample available for the final clustering run in W7–W8.

9.2 Future direction (post-T2 or dependency-blocked)

The following features would materially strengthen the segmentation output but are blocked by frontend instrumentation, platform functionality, or catalogue data that is not available this trimester. Each is documented so it can be added to future iterations without redefining the model.

Deferred feature	Blocked by	Business value when unblocked
outbound_click_rate — clicks leaving DiscountMate for a retailer	Requires Outbound_Merchant_Click event instrumentation. Highest-value single addition.	Turns behavioural analytics from engagement measurement into decision-quality measurement — the outbound click is the closest signal a comparison platform has to a "conversion".
retailer_diversity_index; top_retailer_share	Requires outbound-click instrumentation.	Directly answers "which retailer does this segment prefer?" — the core business question for a comparison platform.

Deferred feature	Blocked by	Business value when unblocked
category_affinity_vector (human-readable)	Requires category ID → category name mapping from Data Engineering product catalogue.	Distinguishes Health Enthusiast, Family Planner, and Premium Shopper in an interpretable way rather than by opaque category IDs.
searches_per_session; filter_usage_rate; sort_usage_rate	Requires search / filter / sort functionality to exist on the platform. /search URL does not yet exist.	Adds a genuine discovery-behaviour dimension currently limited to referrer-based inference.
favourites_count; price_alerts_created; coupon_click_rate	Requires the corresponding platform functionality to be built (favourites, price alerts, coupons). Deferred until App Dev delivers those features.	Enables direct measurement of long-term preference and price-sensitivity.

Outbound-click instrumentation is the single highest-value future addition. It unlocks retailer preference features, provides a genuine purchase-decision signal for a comparison platform, and enables the two-axis loyalty analysis (engagement intensity × decision depth) discussed in Smith (2023). Recommended for the T3 planning cycle as a coordinated App Dev / GA4 tagging item.

10. Success Criteria

Aligned with the roadmap acceptance criteria for DA-01-T3:

- Behavioural feature list completed — 13 features documented in §5.
- Input, cleaning, and output requirements documented — §6.
- Clustering approach defined — §7.
- Downstream contract with DA-02, DA-03, and AI/DL defined — §2 and §6.3.
- Spec reviewed by Shubhpreet (Feature 1 sub-team lead) and shared with Senushi (DA-02 dependency).
- Open decisions closed before extraction (T4) begins — §11.3.

11. Dependencies, Risks, and Open Decisions

11.1 Dependencies

Aligned with the roadmap Section 7.1.

- DA-02-T4 (Senushi) depends on this spec being finalized to lock the dashboard data contract.
- DL-05 (Ethan) depends on cluster output from T11 by W9 for the recommender interface.
- DA-03-T12 (Amoda) optionally depends on persona output for P2 persona-aware scoring.

11.2 Risks

ID	Risk	Mitigation
R1	Data volume — real user volume may remain below the K-means / DBSCAN	Roadmap Risk 1 mitigation: broaden dataset through recruitment, framework-first pipeline validation with

ID	Risk	Mitigation
	meaningful floor even after W6 recruitment.	small samples, and teammate walk-throughs for labelled ground truth.
R2	Category ID mapping delay — without a product-catalogue mapping from Data Engineering, category_browse_ratio cannot be interpreted in human terms in T11.	Raise category name mapping as a cross-team ask with DE this week; fall back to numeric category ID interpretation if unresolved.
R3	Custom-event instrumentation gap — Compare_Page, Specials_Page, etc. currently do not fire on production.	Feature spec is URL-pattern-based, so segmentation is not blocked. Flag to Web Dev for parallel resolution.

11.3 Open decisions

Decisions to close before T4 extraction begins.

ID	Decision	Owner
D1	Confirm minimum inclusion threshold (≥ 1 session with ≥ 10 s engagement)	Shubhpreet
D2	Confirm distance metric/ clustering approach for T9	Shubhpreet (T9)
D3	Confirm category ID $\rightarrow$ category name mapping ownership	Robin $\leftrightarrow$ DE
D4	Confirm outbound-click instrumentation escalation to Web Dev	Robin
D5	Confirm browsing_entropy computation method (Simpson vs Shannon diversity)	Shubhpreet

12. References

Smith, A. (2023). Consumer Behaviour and Analytics. Routledge. The following concepts from Chapters 2 and 3 provide theoretical grounding for this specification:

- The RFE framework adaptation for DVG / comparison platforms (Chapter 2, pp. 33–34) — grounds the choice of framework in §4.
- The attitude–behaviour gap in self-report versus observed data (Chapter 3, pp. 76–77) — grounds the interpretation approach in §4 and §8.
- The systematic-vs-unsystematic behaviour concept (Chapter 2) — grounds the browsing_entropy feature in §5.3.